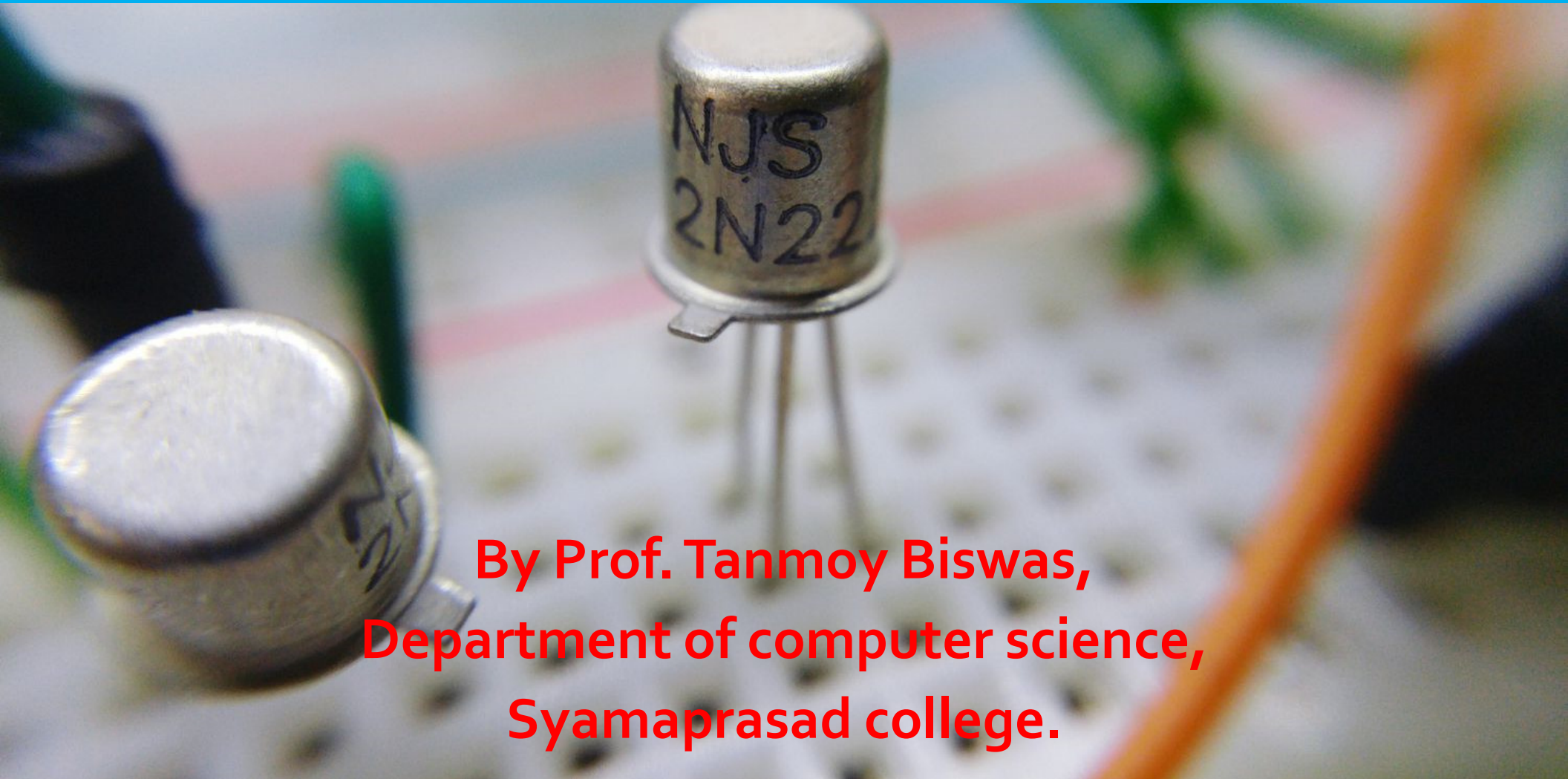


Introduction to TTL Circuits



By Prof. Tanmoy Biswas,
Department of computer science,
Syamaprasad college.

Objective

- Digital Integrated Circuits.
- Bipolar and MOS families.
- TTL – Transistor -Transistor Logic – Characteristics.
- TTL family.
- TTL Parameters.
- Loading rules.

Digital IC Technology

- Digital IC technology started from **Small-Scale Integration (SSI)**, with fewer than 12 gates per chip.
- **Medium-Scale Integration (MSI)**, with 12 to 99 equivalent gates per chip.
- **Large-scale and very large scale integration (LSI and VLSI, respectively)**, which can have tens of thousands of gates per chip.
- **Ultra-large-scale integration (ULSI)**, with over 100,000 gates per chip, and **Giga-scale integration (GSI)**, with 1 million or more.

Digital IC Technology

- ICs have made digital systems more reliable by reducing the external interconnections from one device to another.
- Before IC's, every circuit connection was from one **discrete component** (**transistor**, **diode**, **resistor**, **etc.**) to another.
- Now most of the connections are internal to the ICs, **protected** from **poor soldering, breaks or shorts in connecting paths** on a circuit board, and other physical problems.

Digital IC Technology

- IC's require low **electrical power** to function, **miniature circuitry** requires less power than their discrete counterparts.
- Power supply costs is less, also dissipates less heat and low cooling required.
- ICs are principally used to perform low-power circuit operations that are commonly called ***information processing***.
- The operations that require **high power levels** or **devices** that cannot be integrated are still handled by **discrete components**.

IC logic families.

Bi-polar junction Transistors

- DTL – Diode Transistor Logic.
- TTL – **Transistor-Transistor Logic**.
- ECL – **Emitter Coupled Logic** use bipolar transistors as their major circuit element, also for high speed operations.

MOSFET – **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor.

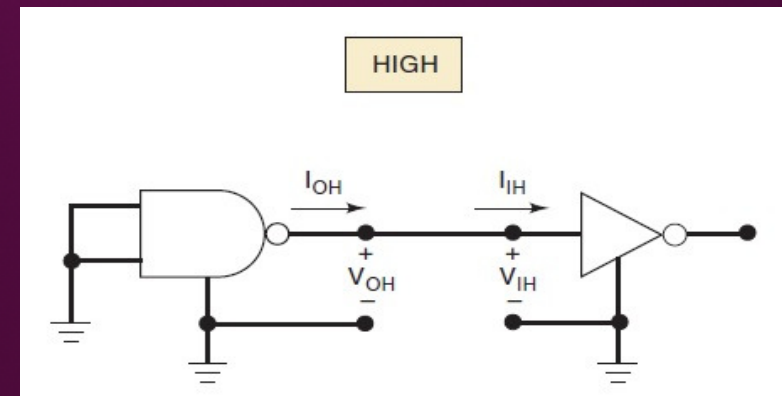
- PMOS.
- NMOS.
- CMOS use unipolar MOSFET transistors as their principal component.

Digital IC Technology

Current and Voltage Parameters

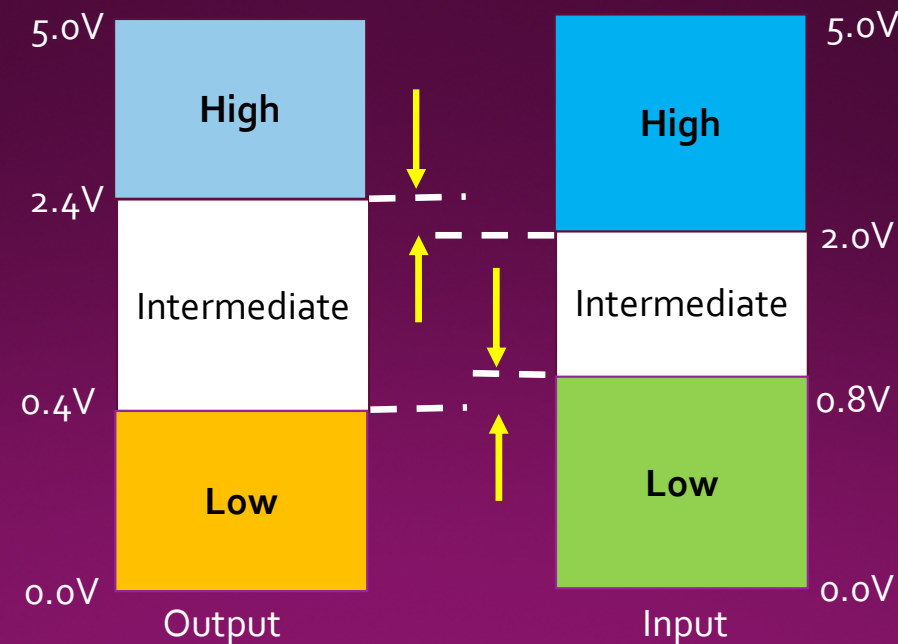
$V_{IH(min)}$ — High-Level Input Voltage. The minimum voltage level required for a logical 1 at an input. Any voltage below this level will not be accepted as a HIGH by the logic circuit.

I_{IH} — High-Level Input Current. The current that flows into an input when a specified high-level voltage is applied to that input.



Driver

Load

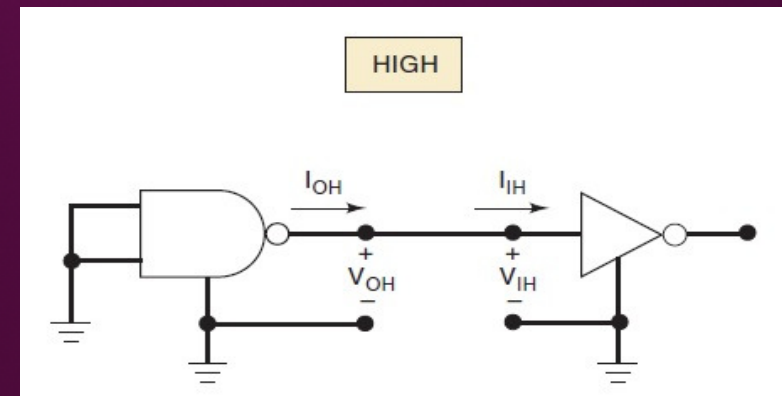


Digital IC Technology

Current and Voltage Parameters

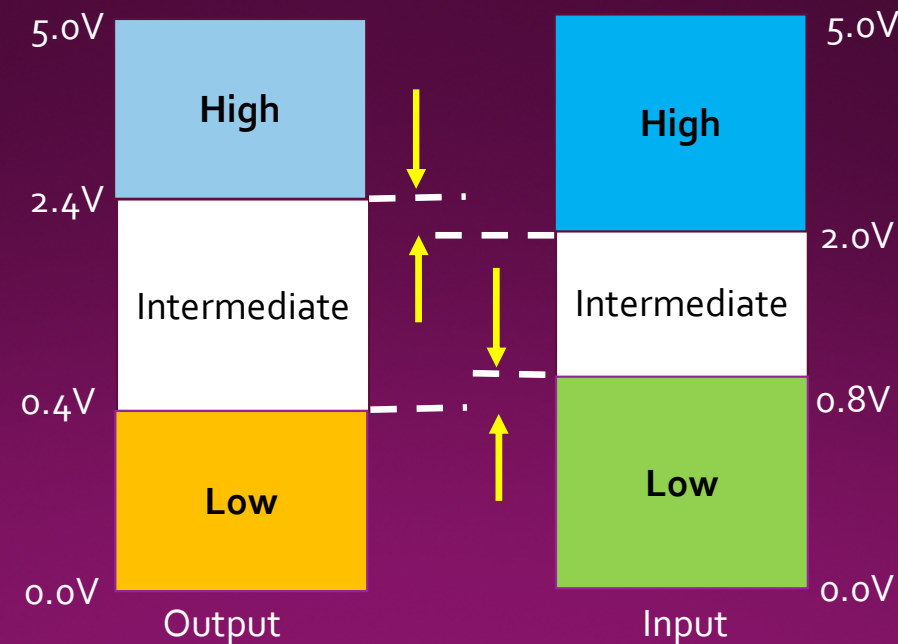
$V_{OH(min)}$ —High-Level Output Voltage. The minimum voltage level at a logic circuit output in the logical 1 state under defined load conditions.

I_{OH} —High-Level Output Current. The current that flows from an output in the logical 1 state under specified load conditions.



Driver

Load

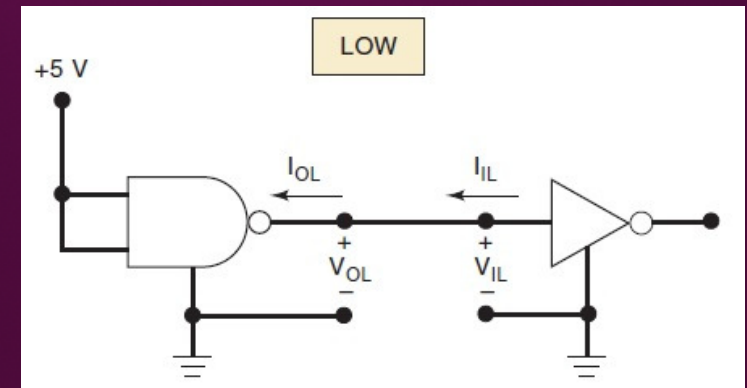


Digital IC Technology

Current and Voltage Parameters

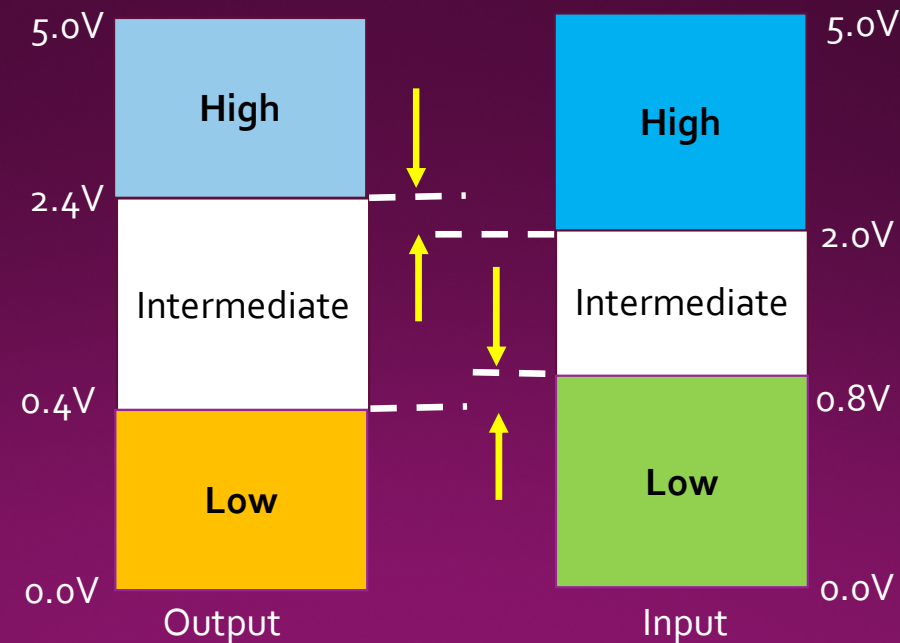
$V_{IL(max)}$ —Low-Level Input Voltage. The maximum voltage level required for a logic 0 at an input. Any voltage above this level will not be accepted as a LOW by the logic circuit.

I_{IL} —Low-Level Input Current. The current that flows into an input when a specified low-level voltage is applied to that input.



Driver

Load

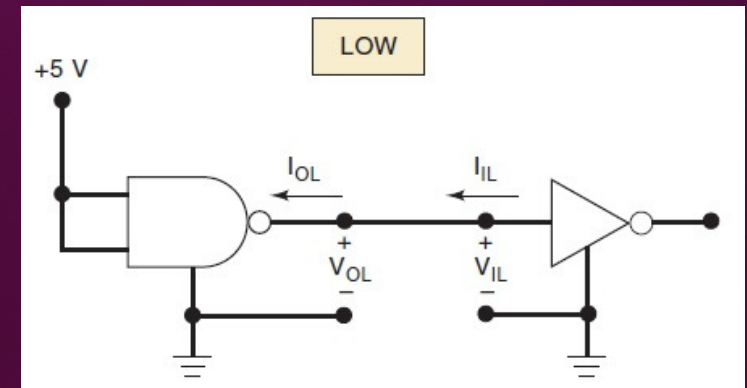


Digital IC Technology

Current and Voltage Parameters

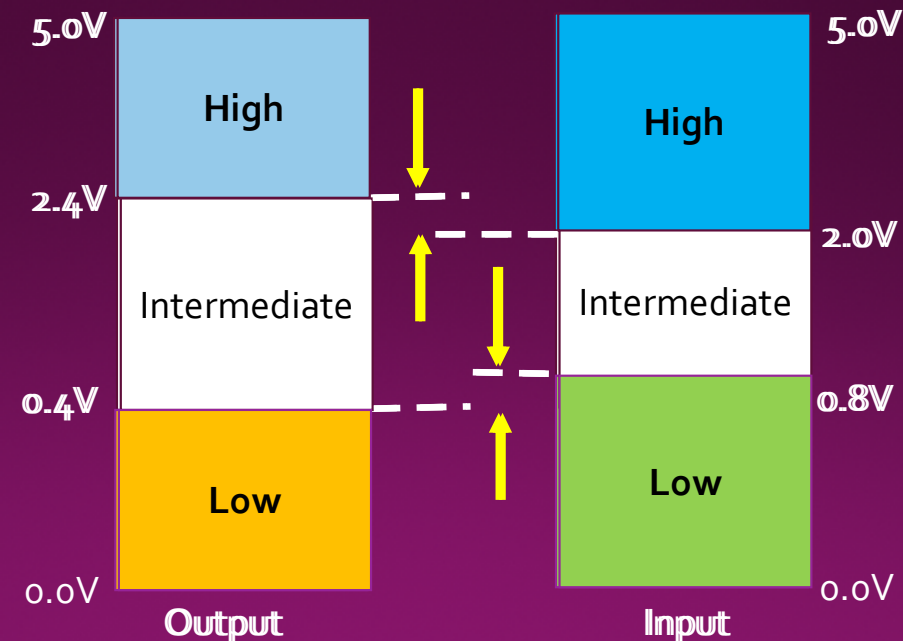
$V_{OL(max)}$ —Low-Level Output Voltage. The maximum voltage level at a logic circuit output in the logical 0 state under defined load conditions.

I_{OL} —Low-Level Output Current. The current that flows from an output in the logical 0 state under specified load conditions.



Driver

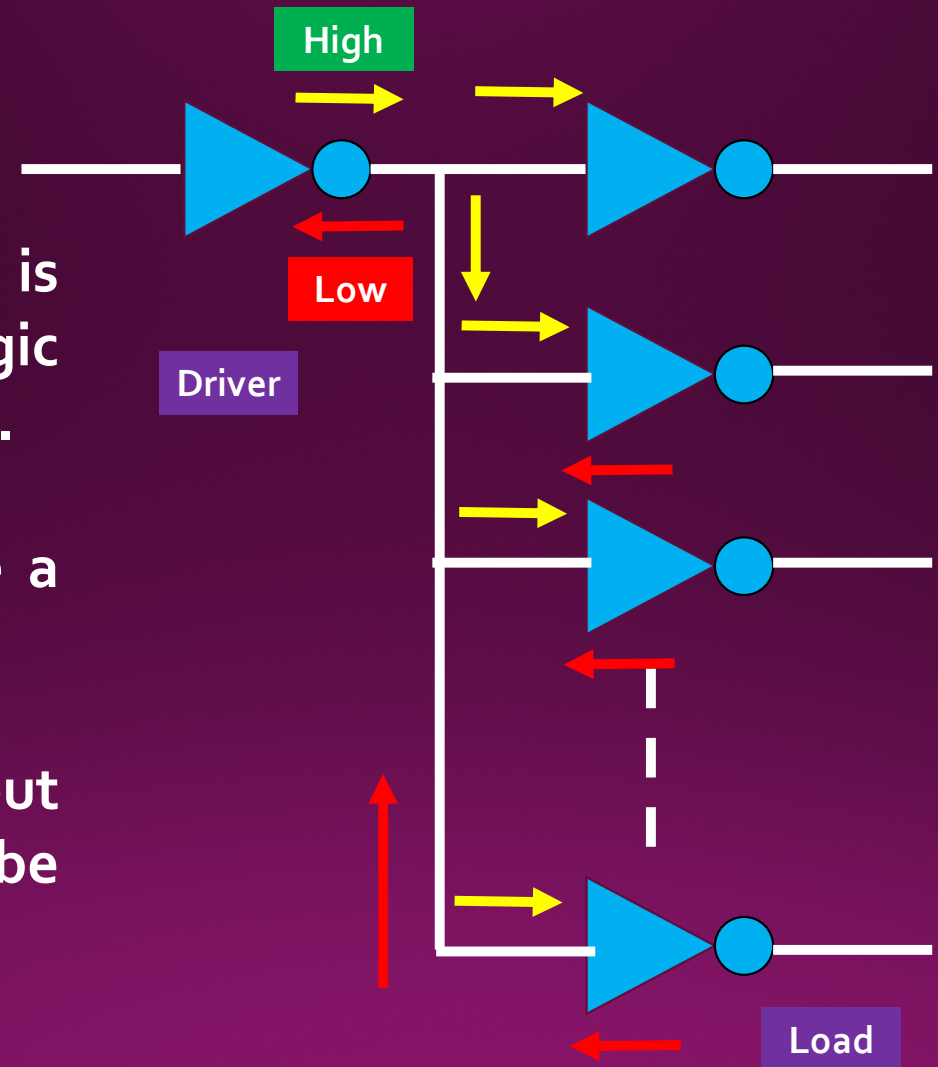
Load



Digital IC Technology

Fan-Out

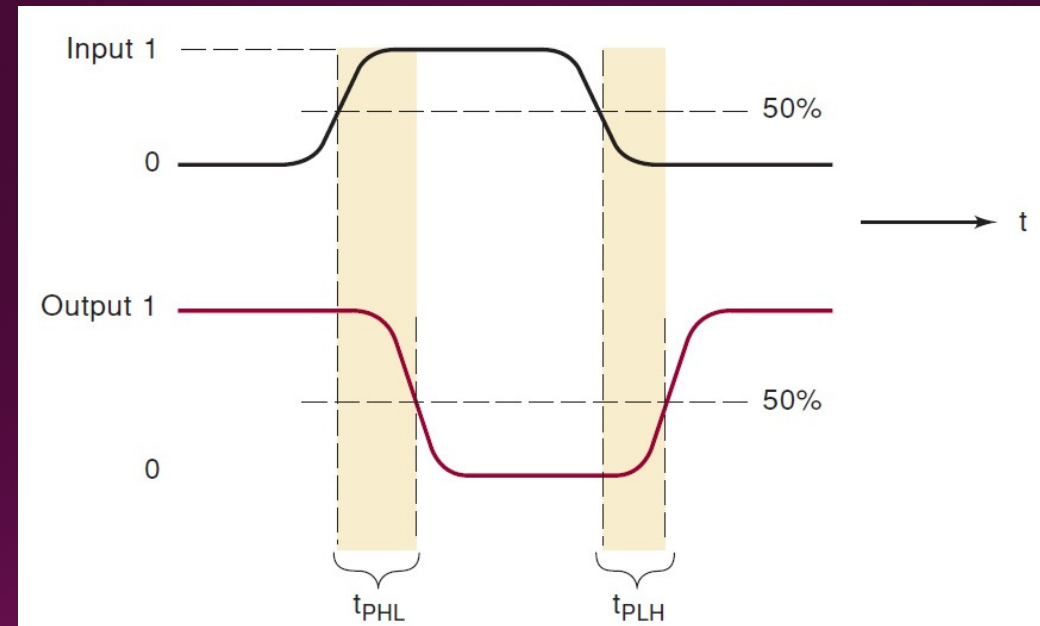
- The fan-out (also called **loading factor**) is defined as the maximum number of logic inputs that an output can drive reliably.
- A logic gate that is specified to have a fan-out of 10 can drive 10 logic inputs.
- If this number is exceeded, the output logic-level voltages cannot be guaranteed.



Digital IC Technology

Propagation Delay

A logic signal always experiences a delay in going through a circuit. The two propagation delay times are defined as follows:

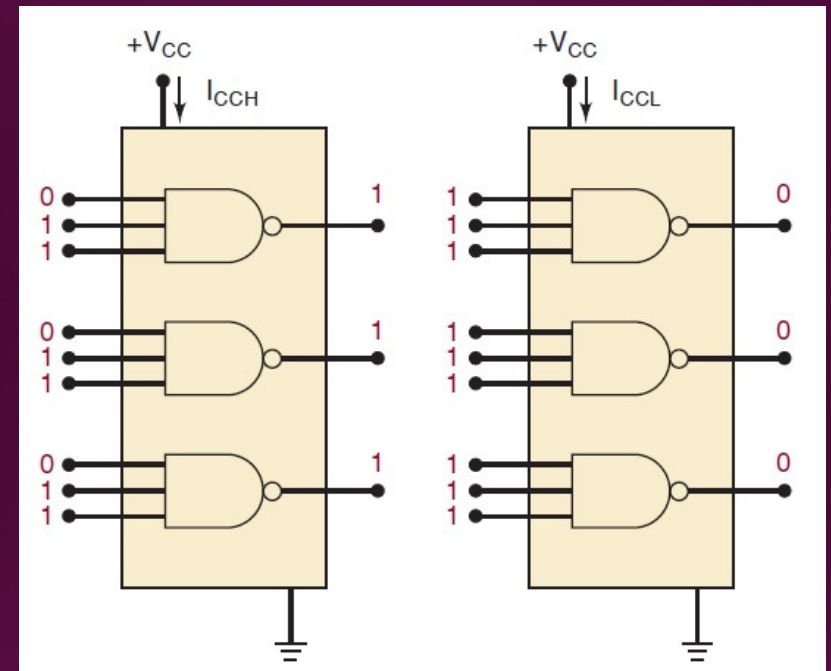


- t_{PLH} : Delay time in going from **logical 0** to **logical 1** state (LOW to HIGH).
- t_{PHL} : Delay time in going from **logical 1** to **logical 0** state (HIGH to LOW).

Digital IC Technology

Power Requirement

Every IC requires a certain amount of electrical power to operate. This power is supplied by one or more power-supply voltages connected to the power pin(s) on the chip labeled V_{CC} (for TTL) or V_{DD} (for MOS devices).

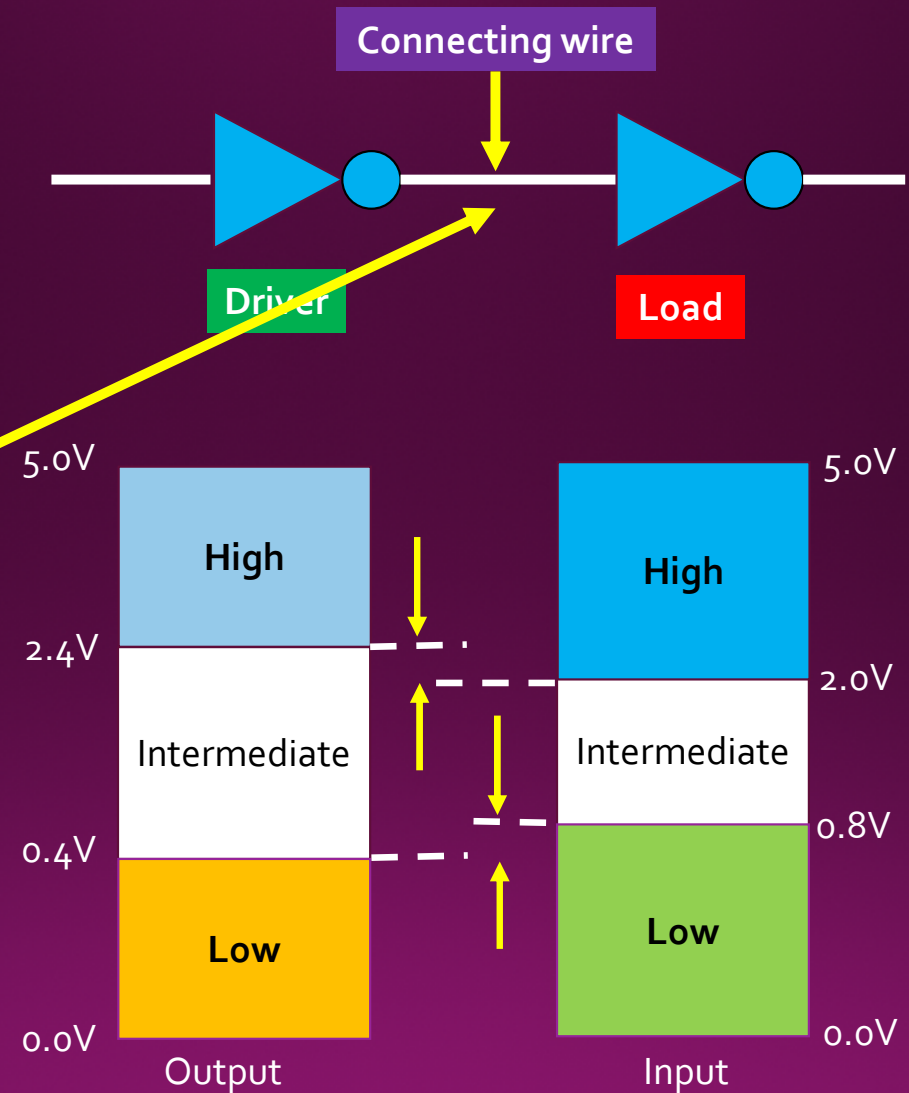


$$I_{CC}(\text{avg}) = \frac{I_{CCH} + I_{CCL}}{2}$$

$$P_D(\text{avg}) = I_{CC}(\text{avg}) \times V_{CC}$$

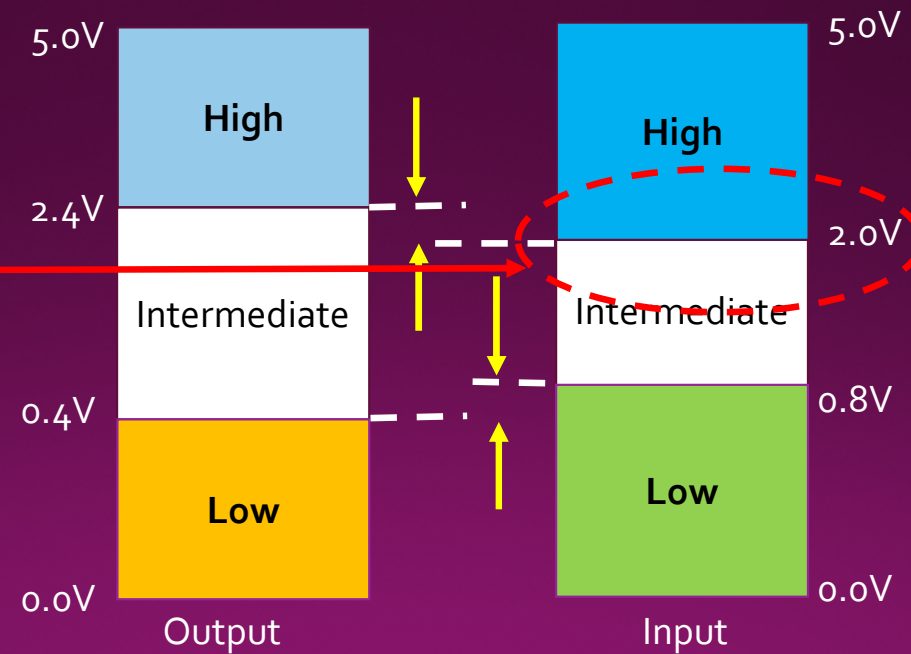
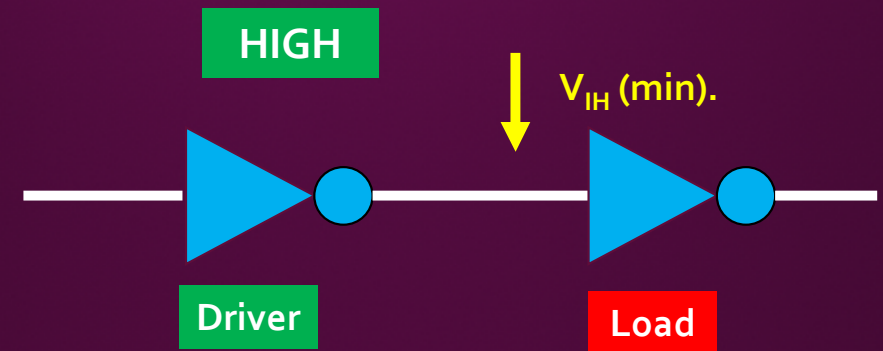
Noise Immunity

Stray electric and magnetic fields can induce voltages on the connecting wires between logic circuits are usually called **noise**.



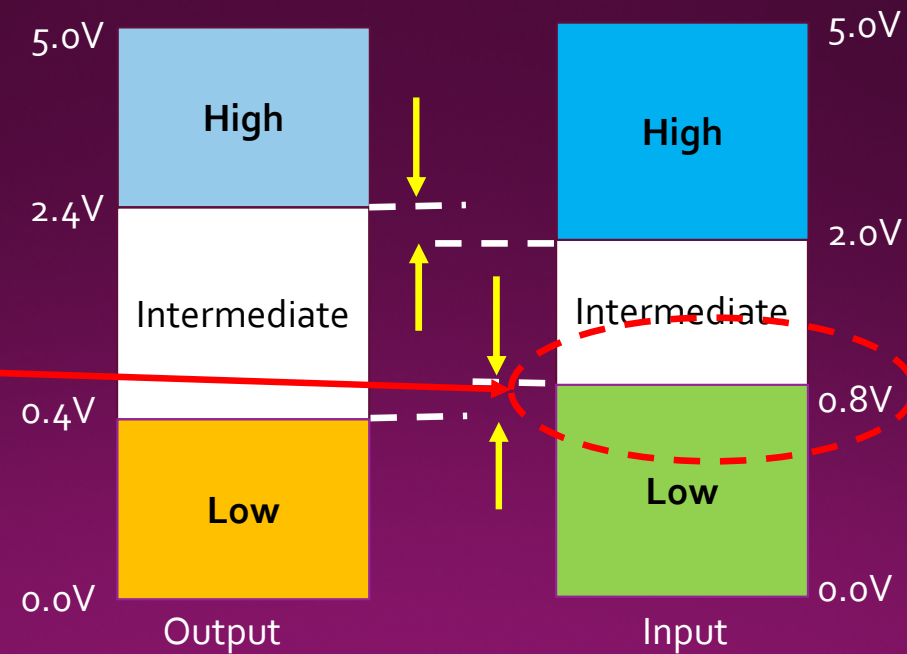
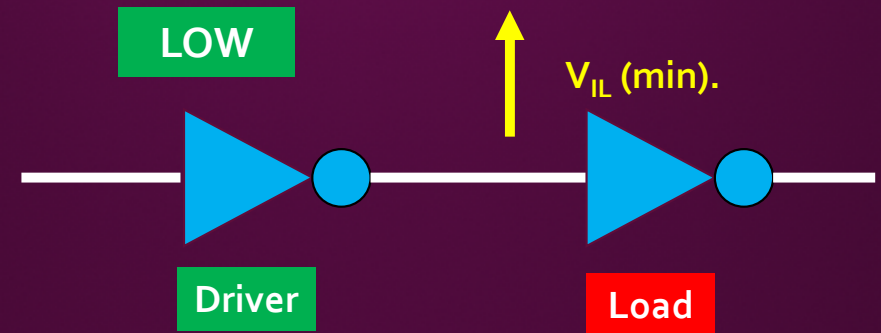
Noise Immunity

These are called **noise** and can cause the voltage at the input to a logic circuit to drop below V_{IH} (min).

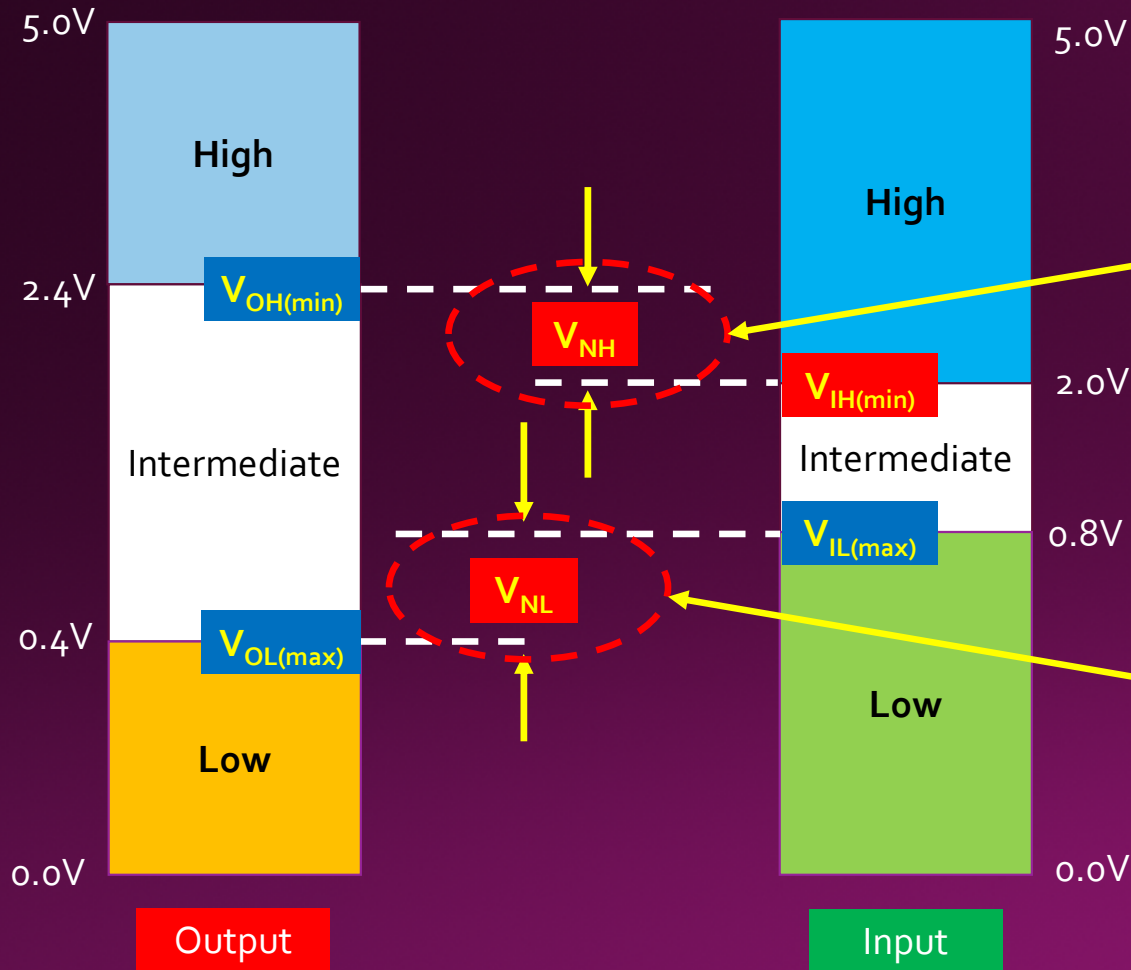


Noise Immunity

It can also increase the input voltage at low logic $V_{IL}(\text{max})$, which could produce unpredictable operation.



Noise Immunity



High-state noise margin

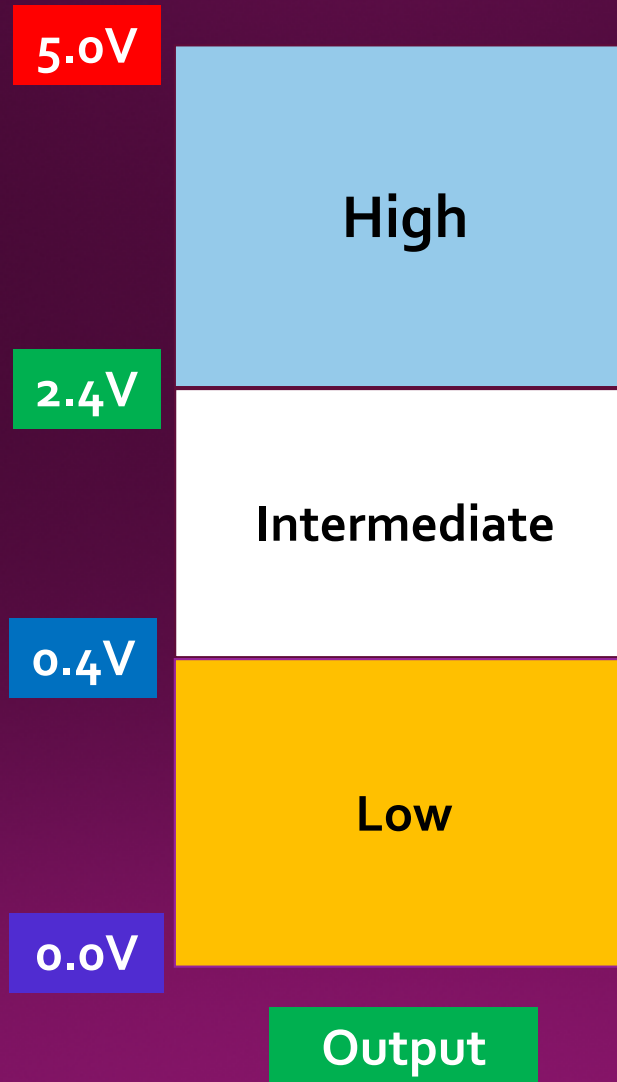
$$V_{NH} = V_{OH(min)} - V_{IH(min)}$$

Low-state noise margin

$$V_{NL} = V_{IL(max)} - V_{OL(max)}$$

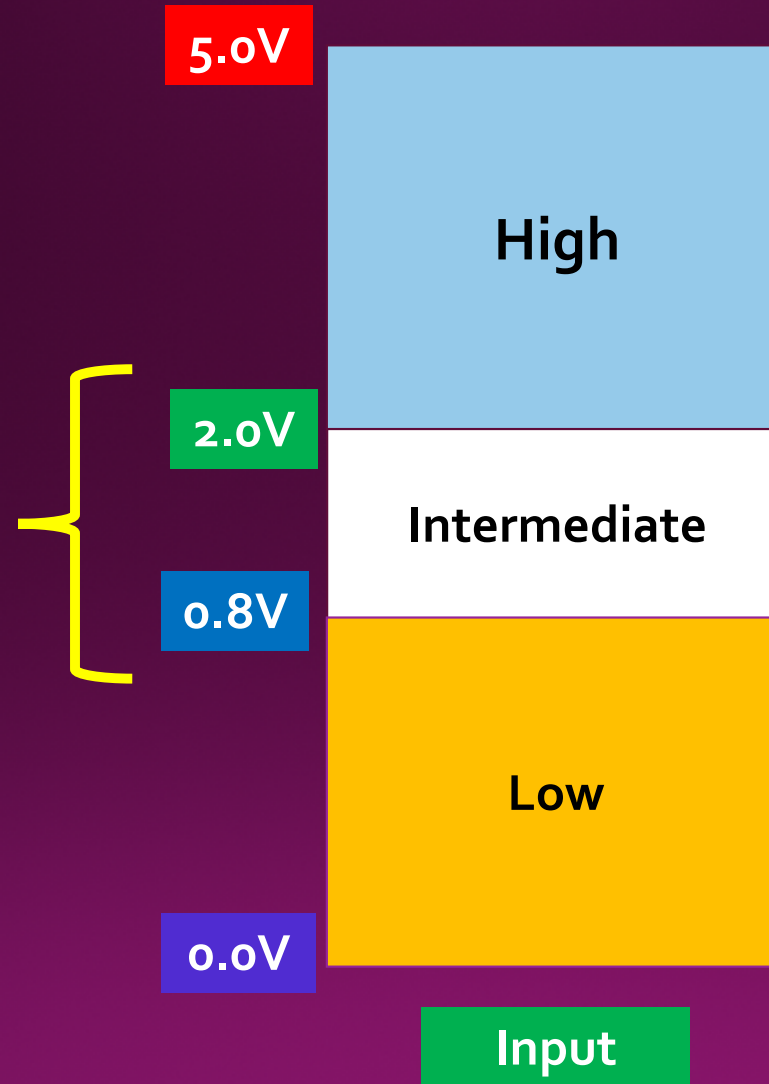
Invalid Voltages

Output Profile of TTL –
Invalid voltage ranges are
between 0.4V to 2.4V.

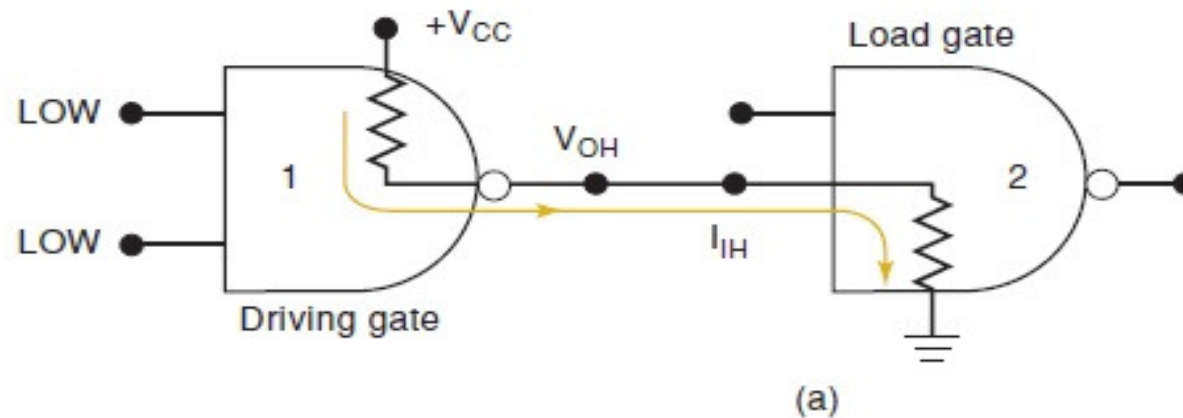


Invalid Voltages

Input Profile of TTL – Invalid voltage ranges are between 0.8V to 2.0V.

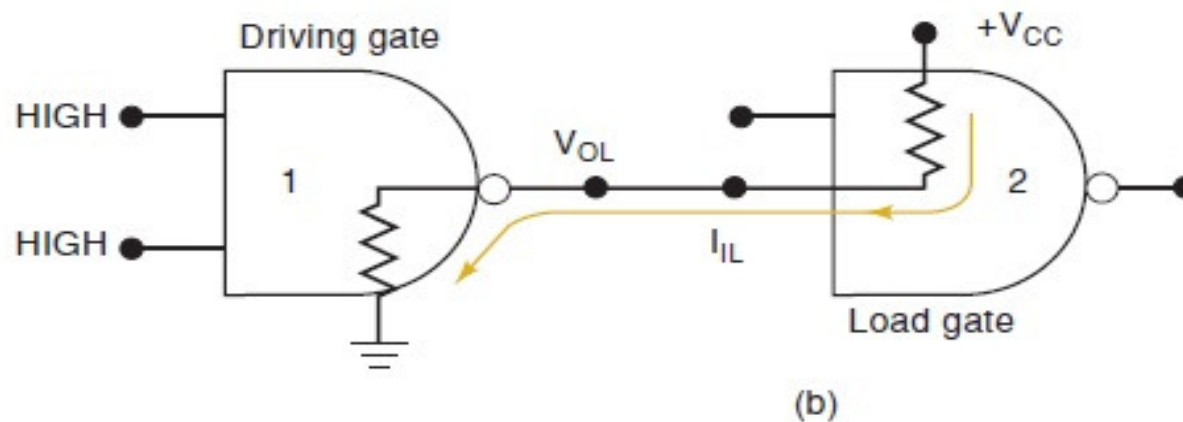


Current Sinking and Sourcing



Current sourcing

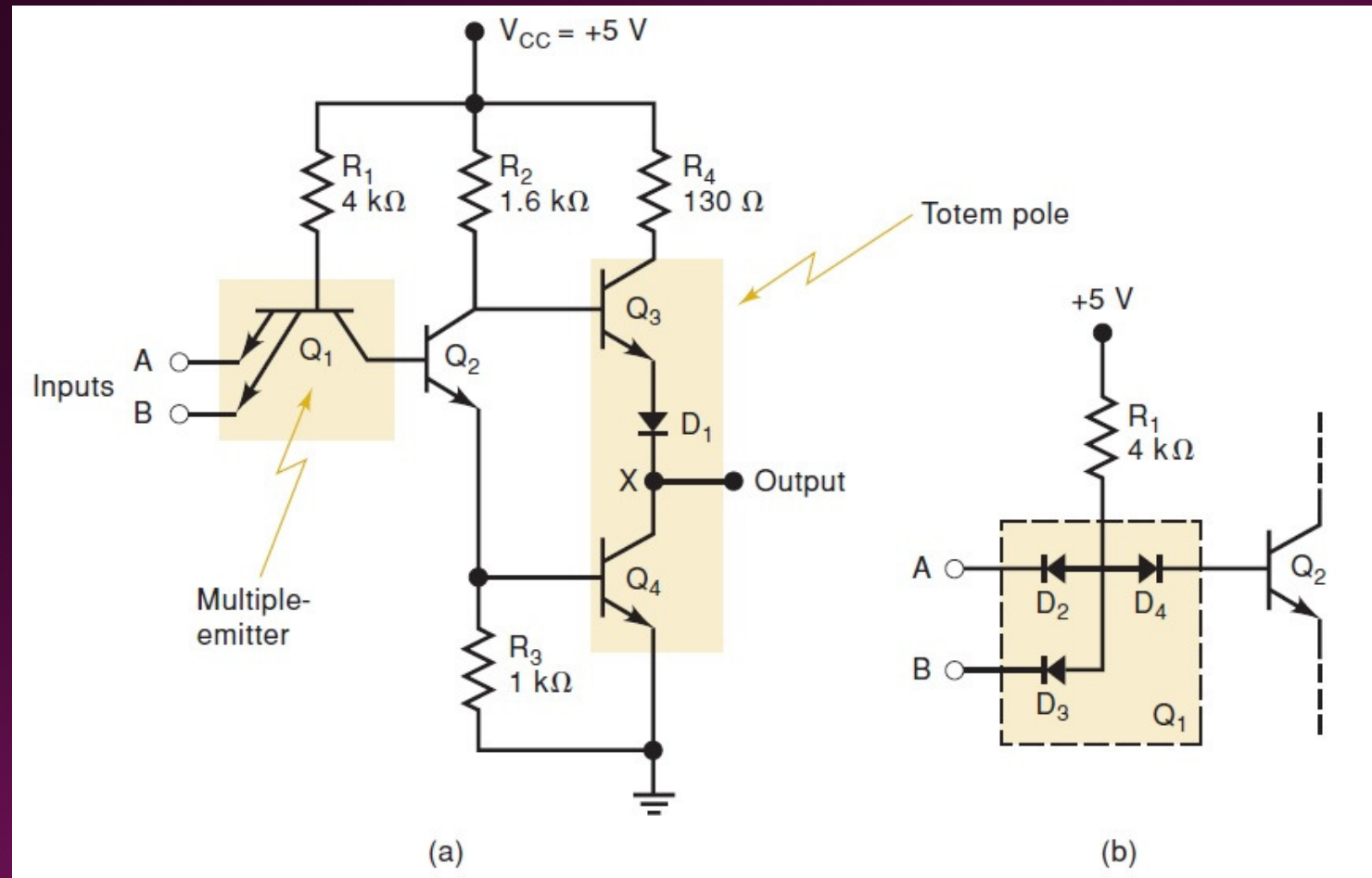
Driving gate supplies (sources) current to load gate in HIGH state.



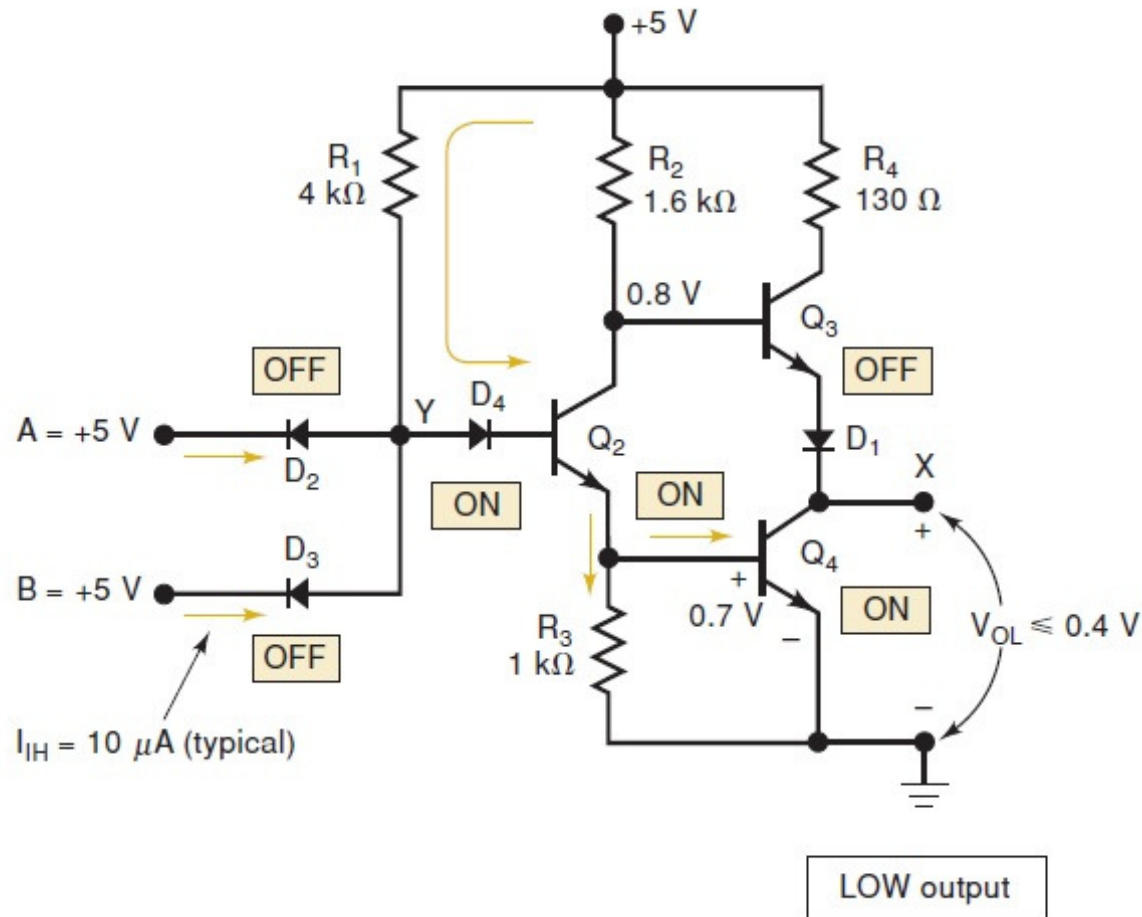
Current sinking

Driving gate receives (sinks) current from load gate in LOW state.

The TTL Logic family – Standard TTL

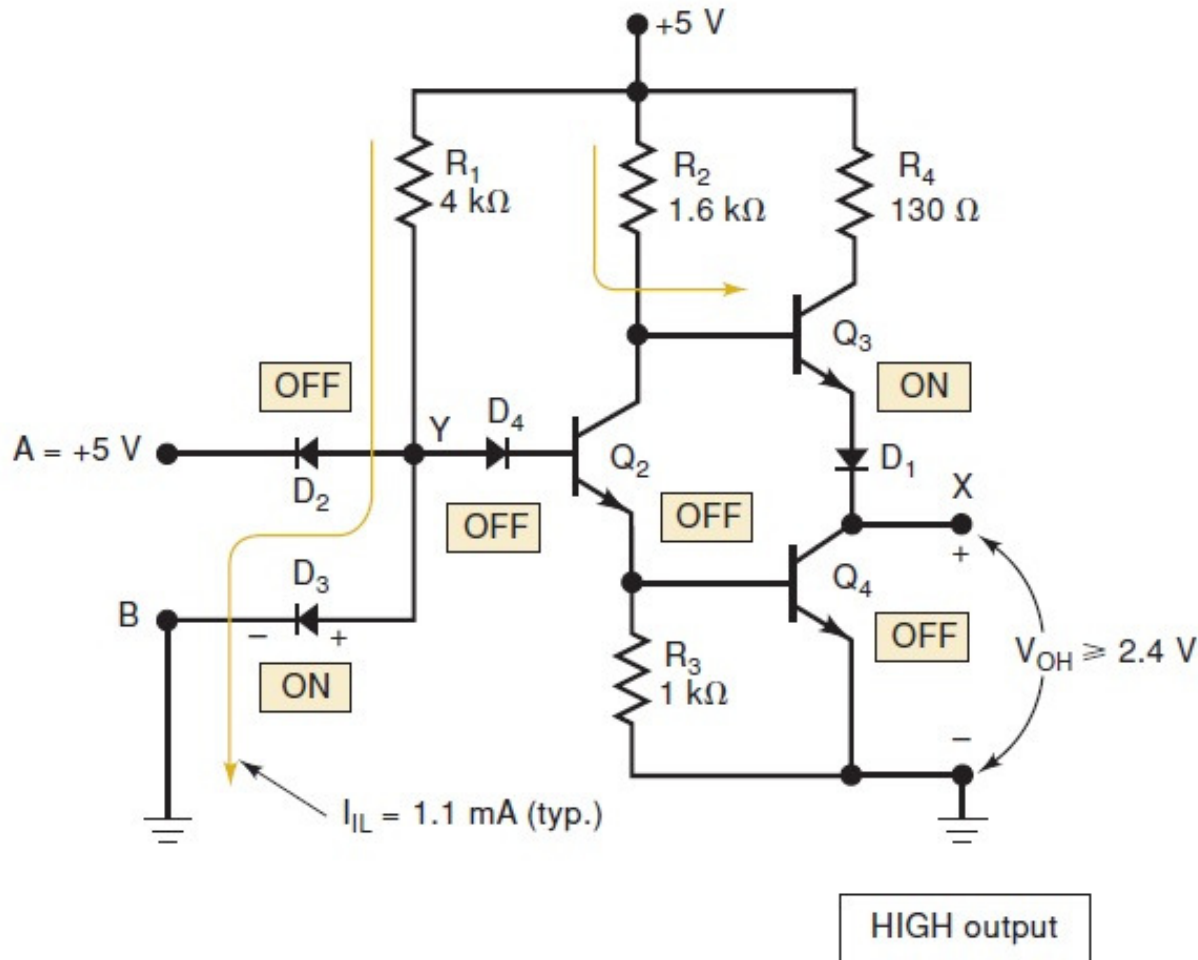


Circuit Operation—LOW output State



Input conditions	Output conditions
A and B are both HIGH ($\geq 2\text{ V}$)	Q_3 OFF
Input currents are very low $I_{IH} = 10\text{ }\mu\text{A}$	Q_4 ON so that V_X is LOW ($\leq 0.4\text{ V}$)

Circuit Operation—High output State



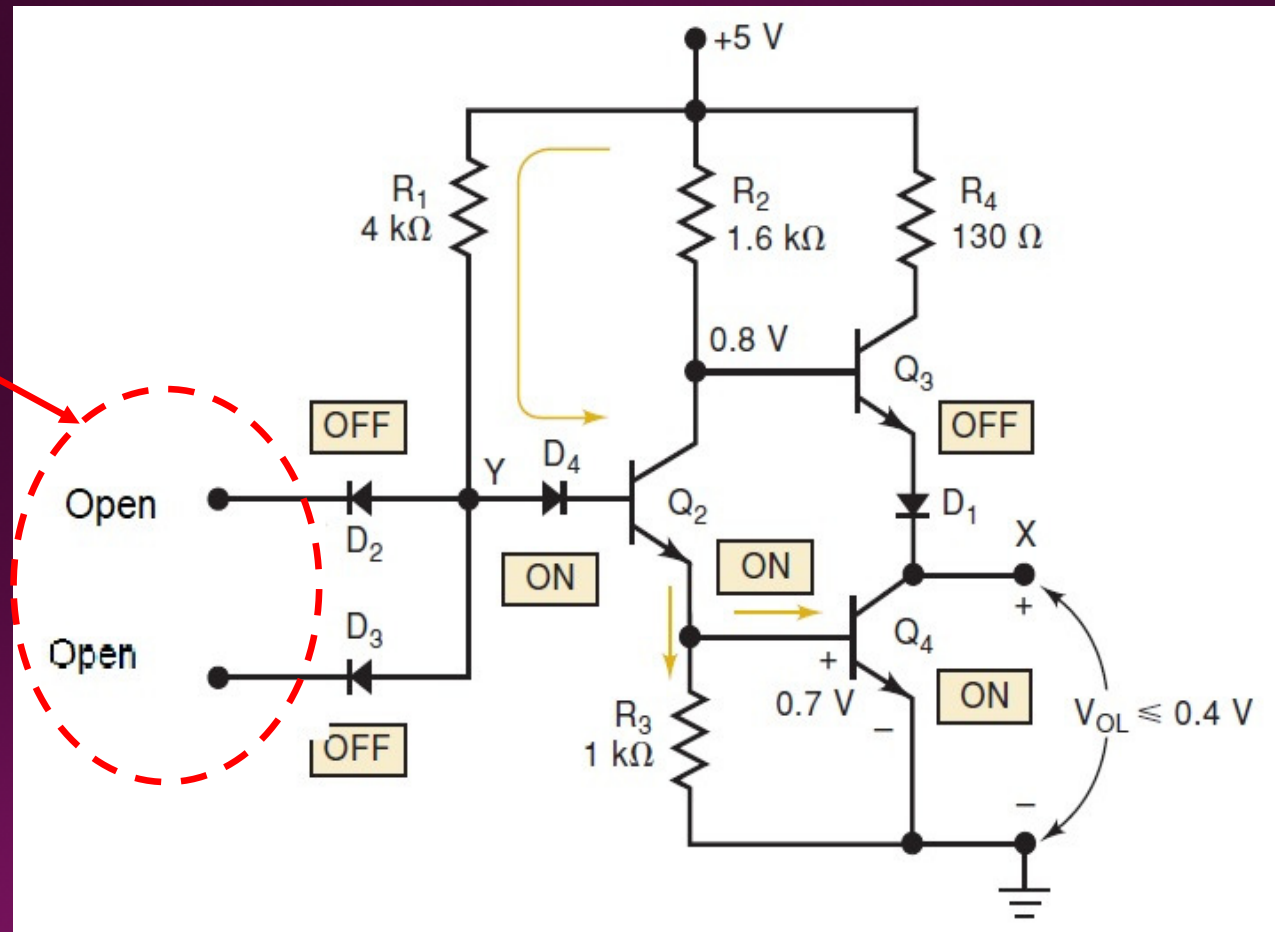
Input conditions	Output conditions
A or B or both are LOW (≤ 0.8 V)	Q_4 OFF
Current flows back to ground through LOW input terminal. $I_{IL} = 1.1$ mA	Q_3 acts as emitter-follower and $V_{OH} \geq 2.4$ V, typically 3.6 V

Circuit Operation—when both the inputs are open

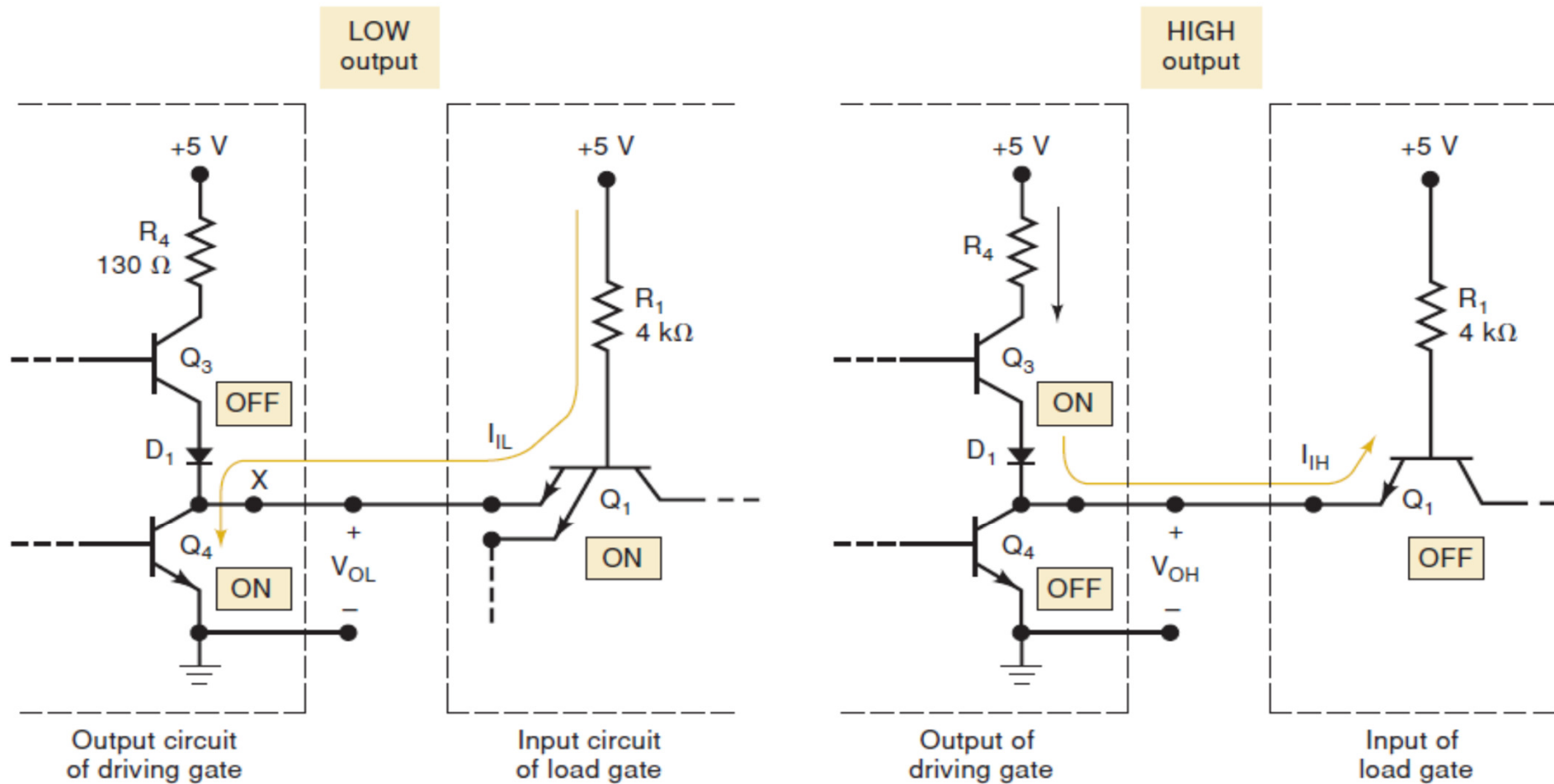
Called floating inputs

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Truth table of NAND gate



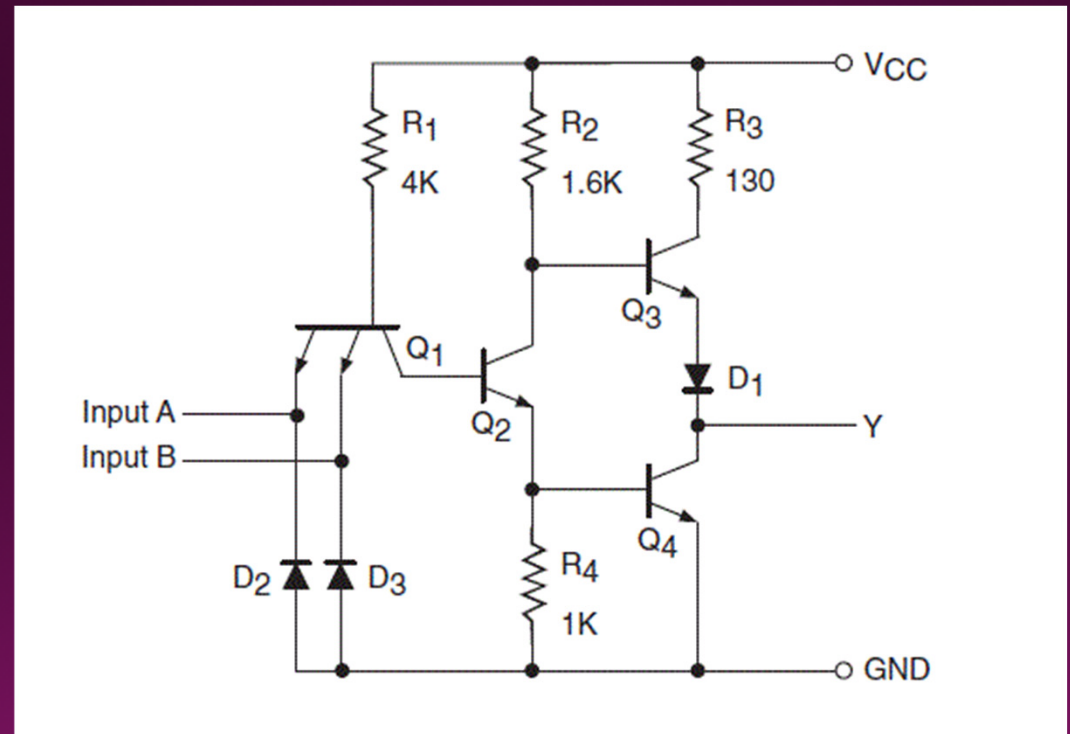
Circuit Operation—Current Sourcing and Sinking



Transistor-Transistor Logic & Family.

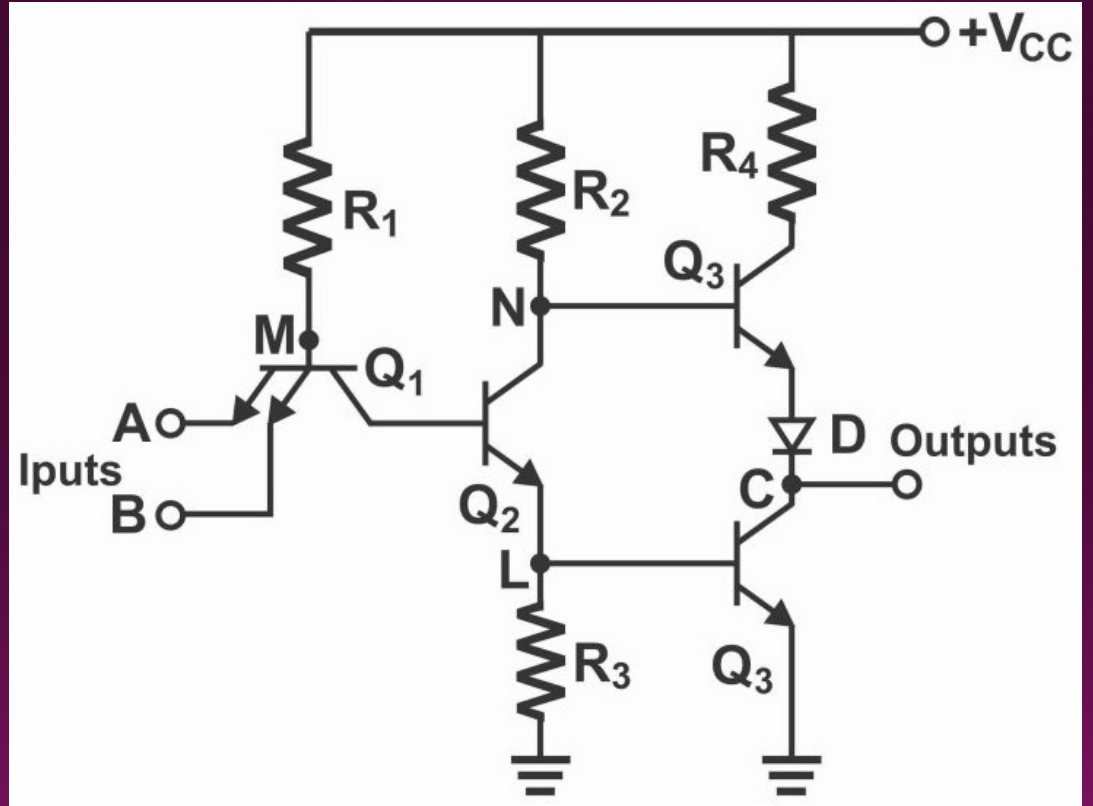
TTL SERIES CHARACTERISTICS

- Standard TTL is also called High Power/Speed TTL.
- Power dissipation – **22mW**.
- Propagation delay is **6nsec**.
- Usually denoted as **74HXX**.



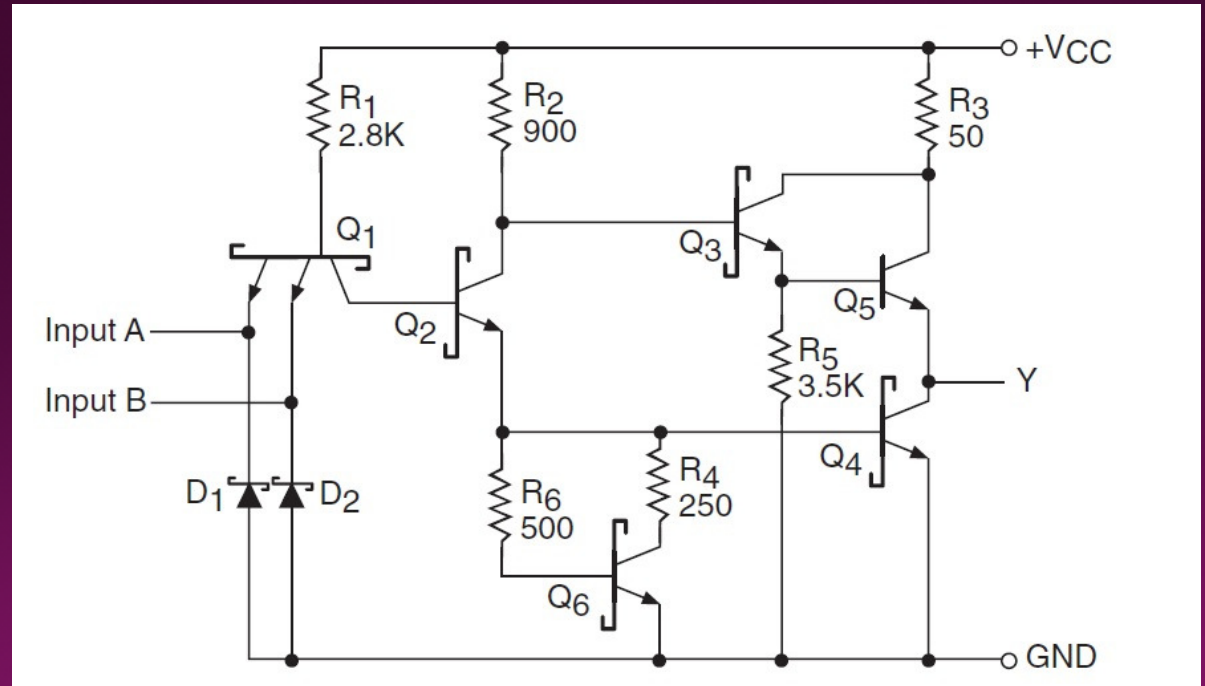
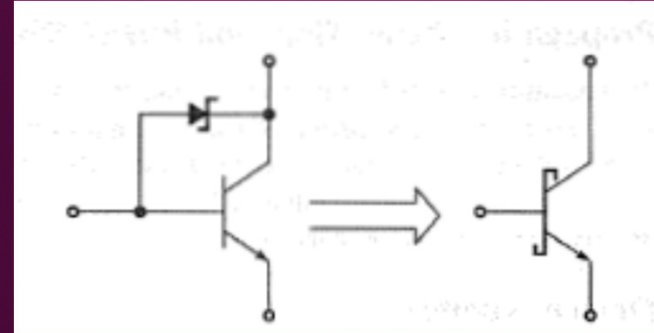
Low Power /Speed TTL

- **Low Power TTL.**
- **Power dissipation – 1mW.**
- **Propagation delay is 35nsec.**
- **Usually denoted as 74LXX**



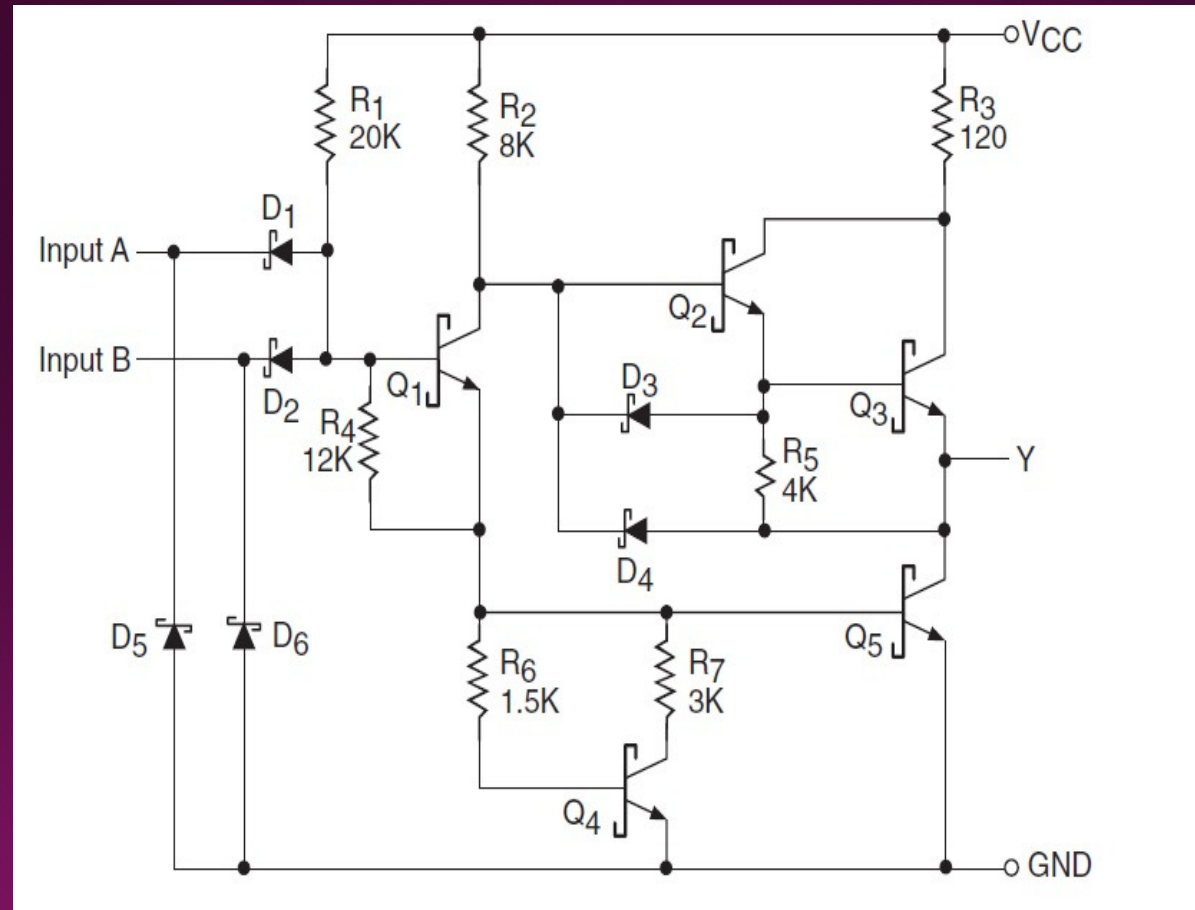
Schottky TTL

- Are very fast – **100MHz**.
- Low Power TTL.
- Power dissipation – **20mW**.
- Propagation delay is **3nsec**.
- Usually denoted as 74SXX.

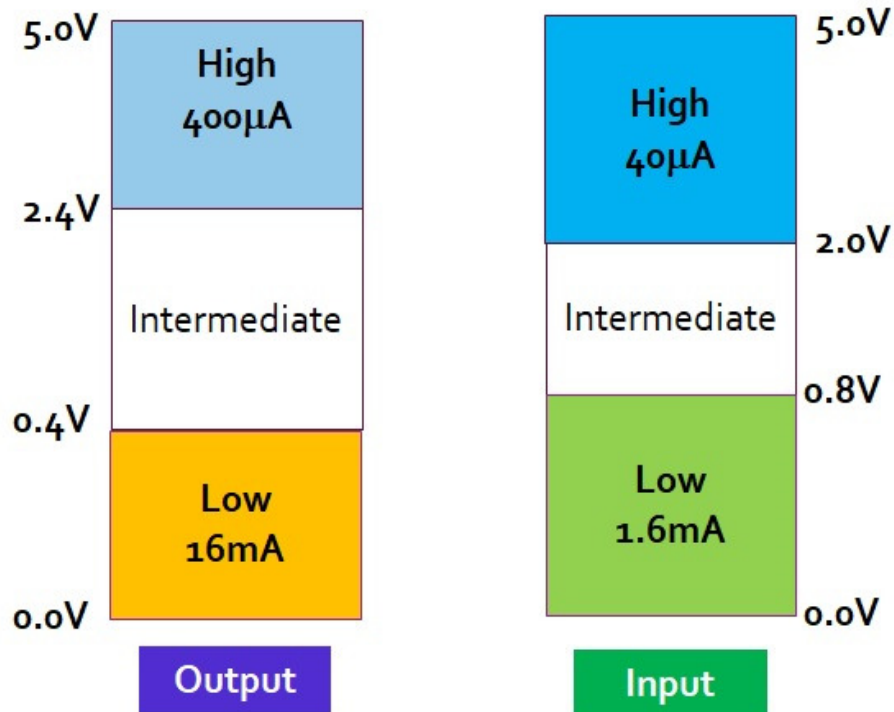


Low Power Schottky TTL

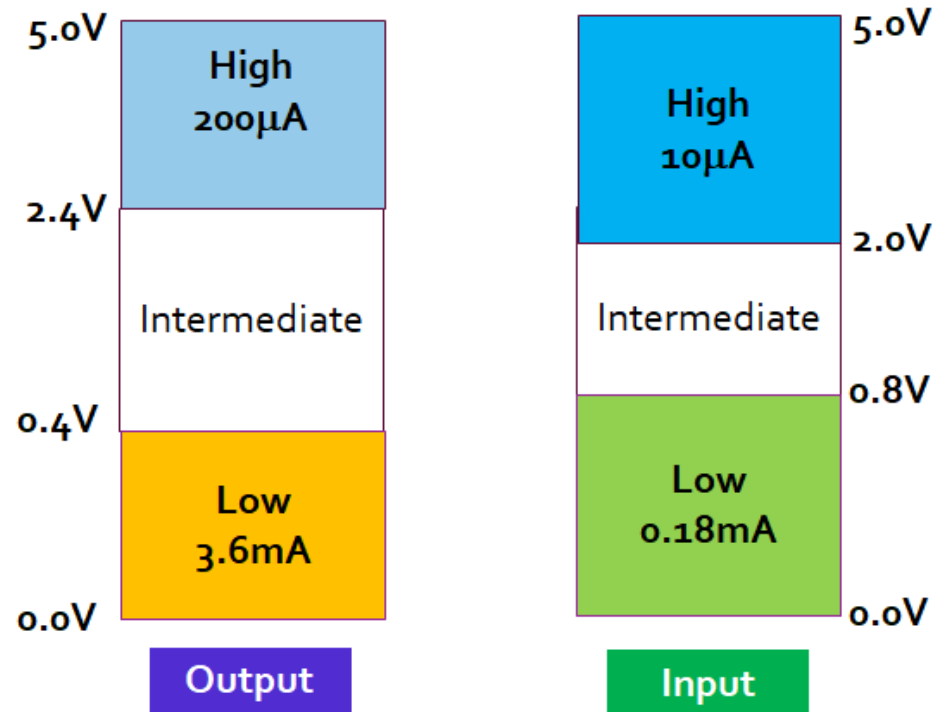
- Low Power TTL.
- Power dissipation – 2mW.
- Propagation delay is 10nsec.
- Usually denoted as 74LSXX.



Loading rules – Fan-Out

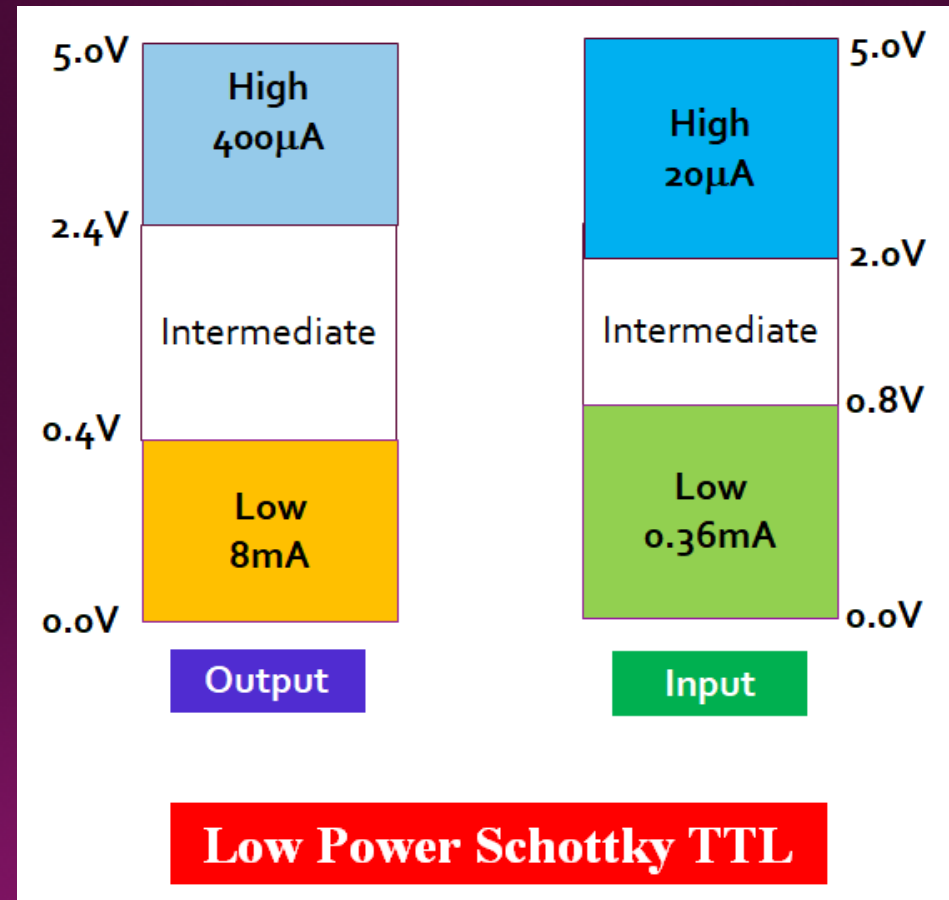
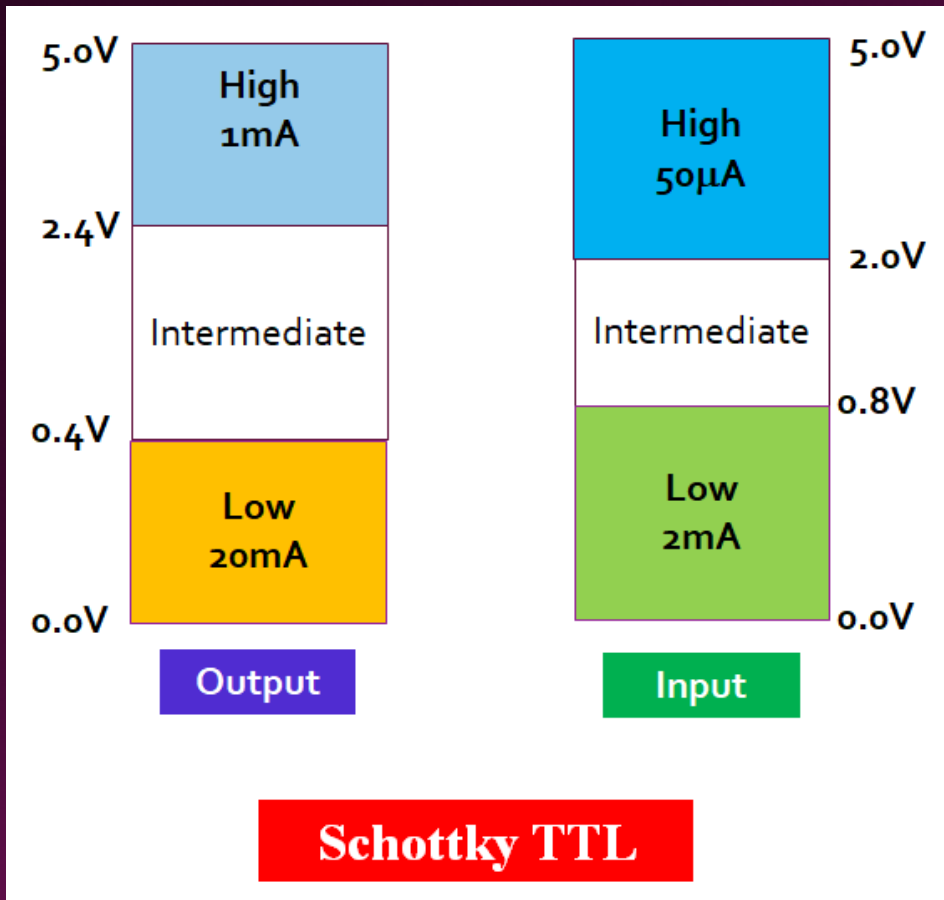


Standard TTL

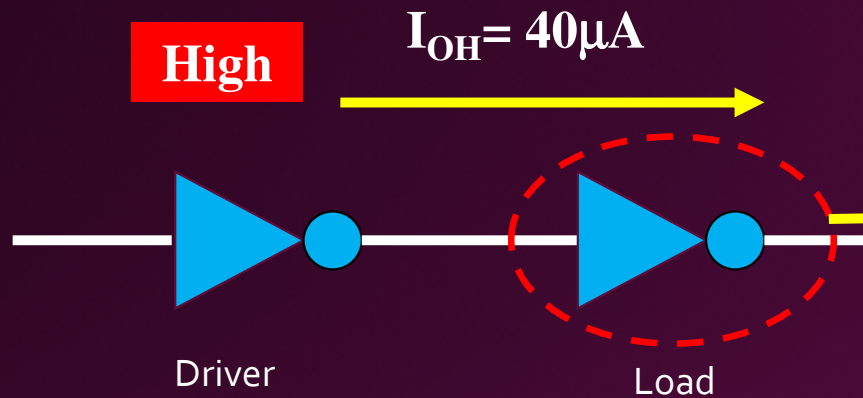


Low Power TTL

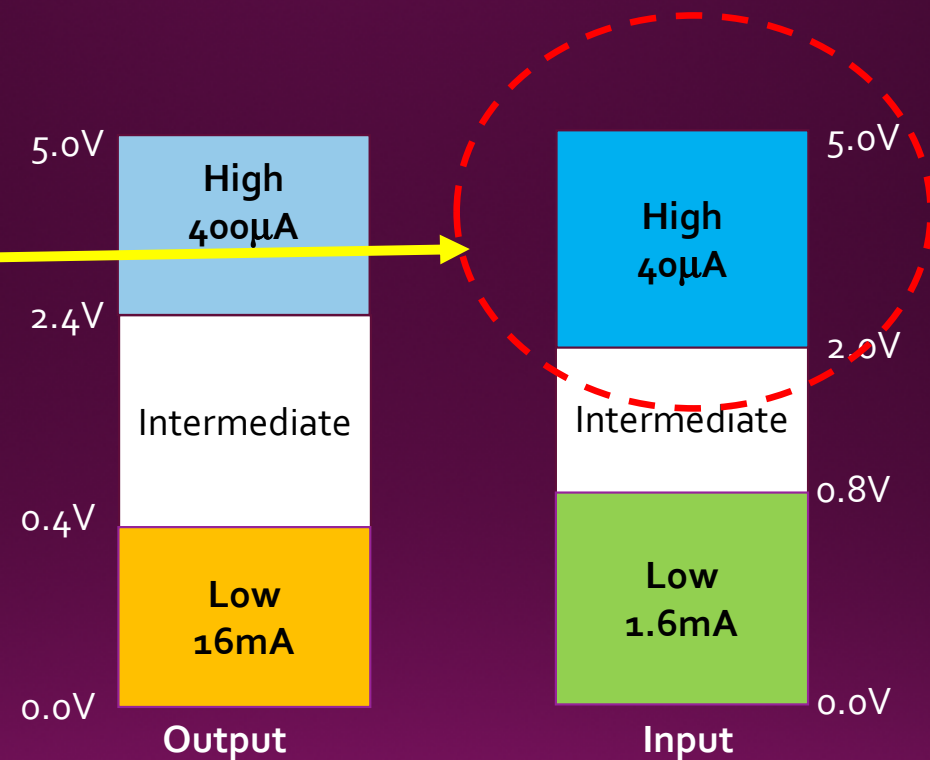
Loading rules – Fan-Out



Loading rules – Fan-Out

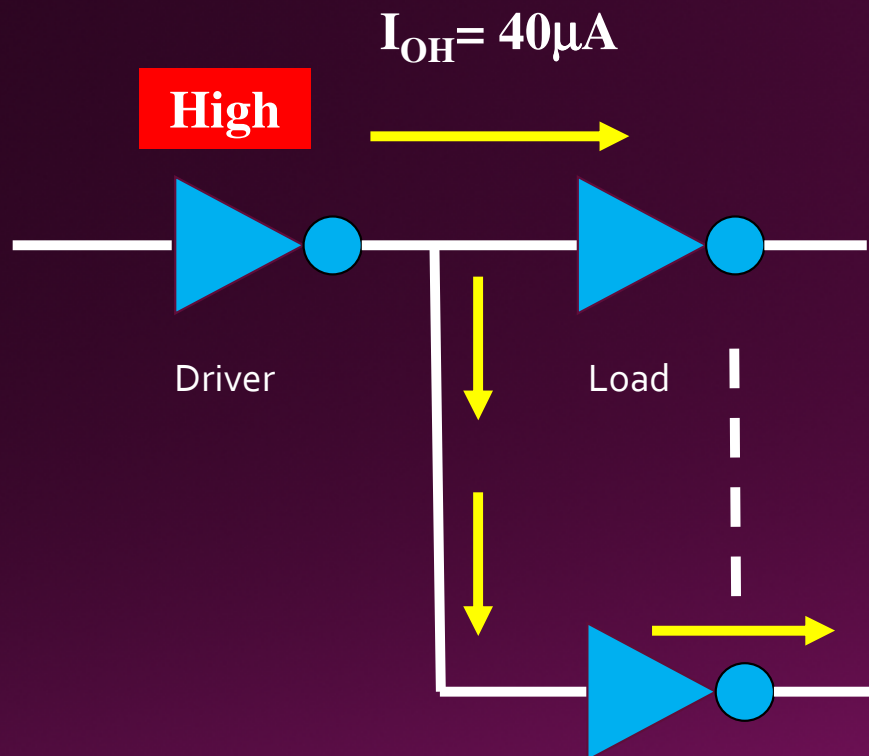


- When the drive output is high it will source out current and it will be equal to current needed by the Load TTL gate.
- As per the input profile the TTL gate requires only 40mA.

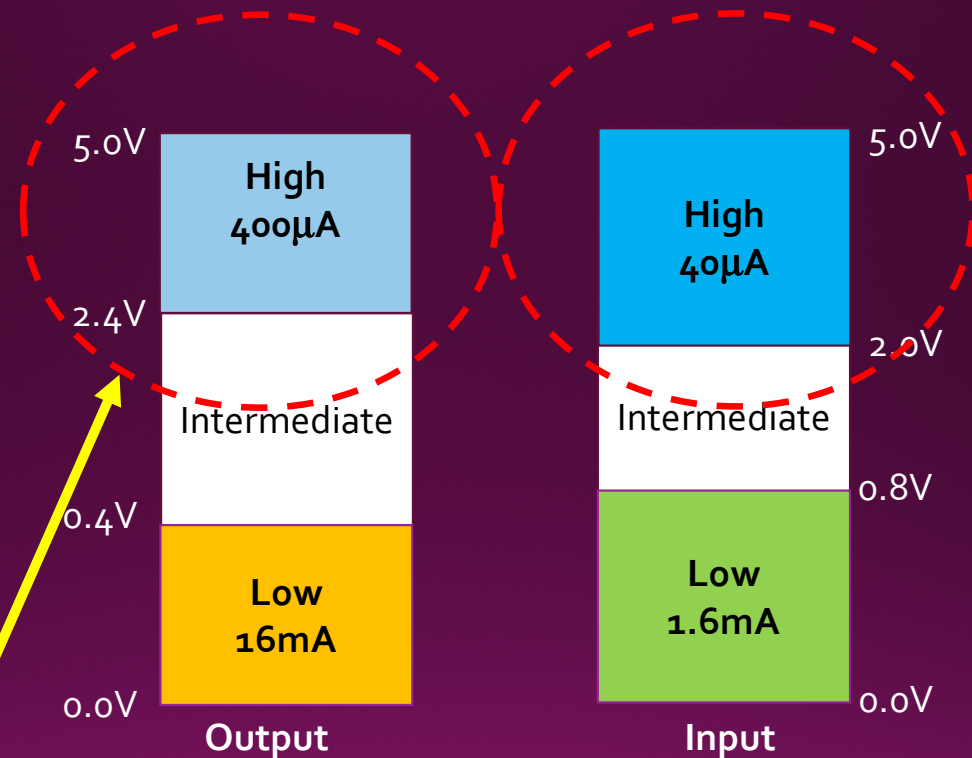


Standard TTL

Loading rules – Fan-Out

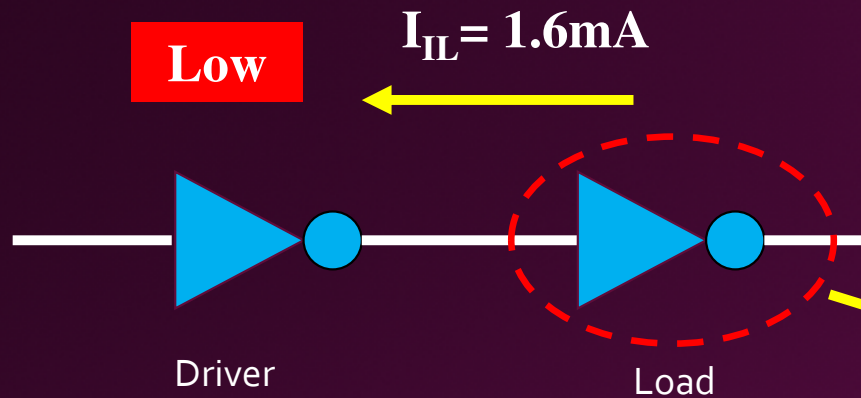


$$I_{OH(max)} = 40\mu A \times 10 = 400\mu A$$

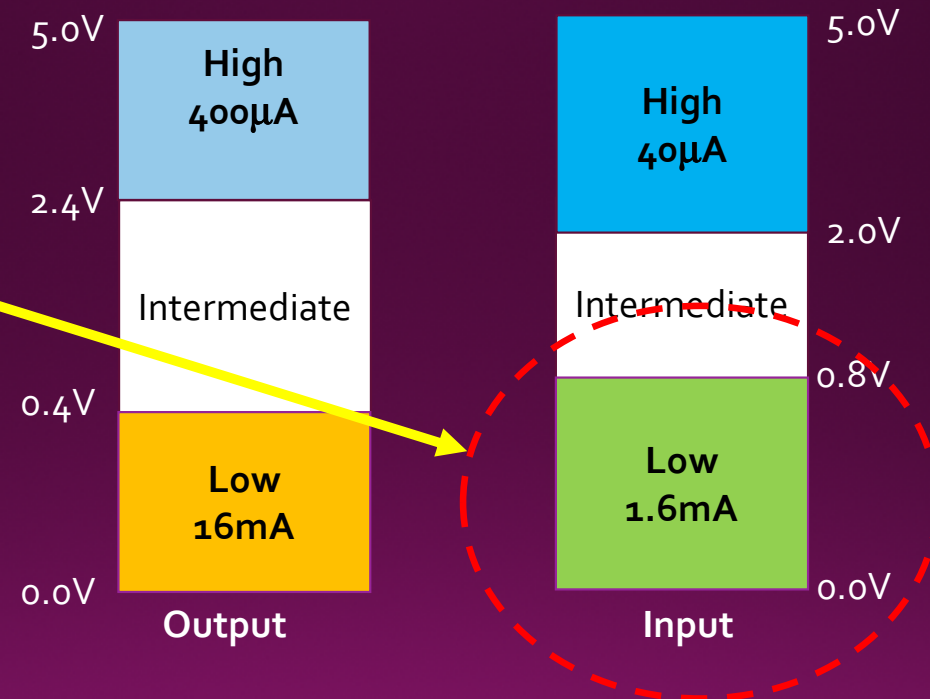


Standard TTL

Loading rules – Fan-Out

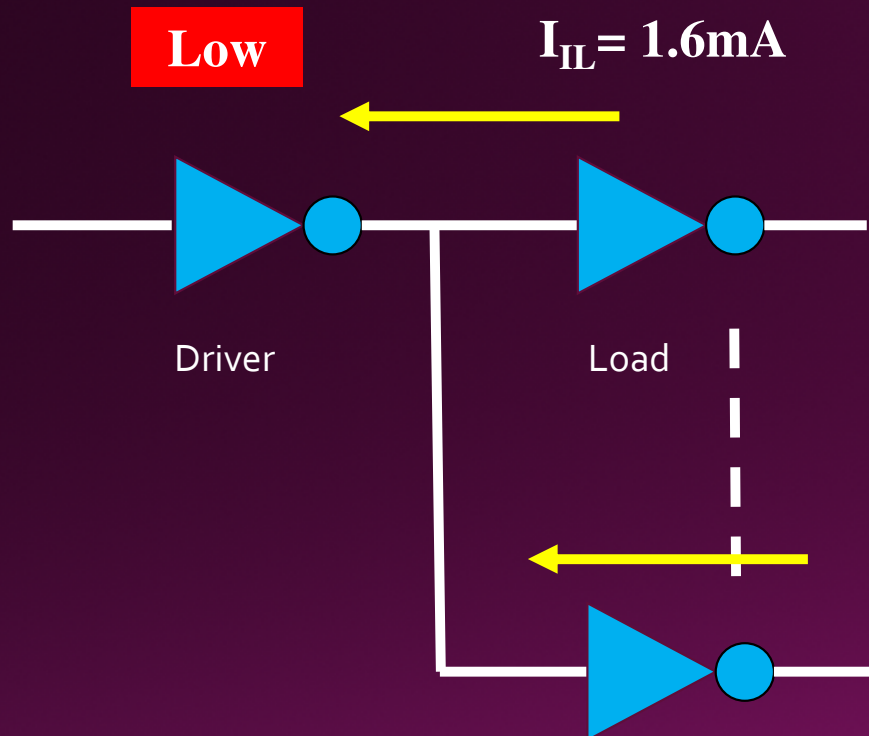


- When the drive output is low it will sink in current and it will be equal to current sourced by one Load TTL gate.
- As per the input profile the TTL load gate will source only 1.6mA.

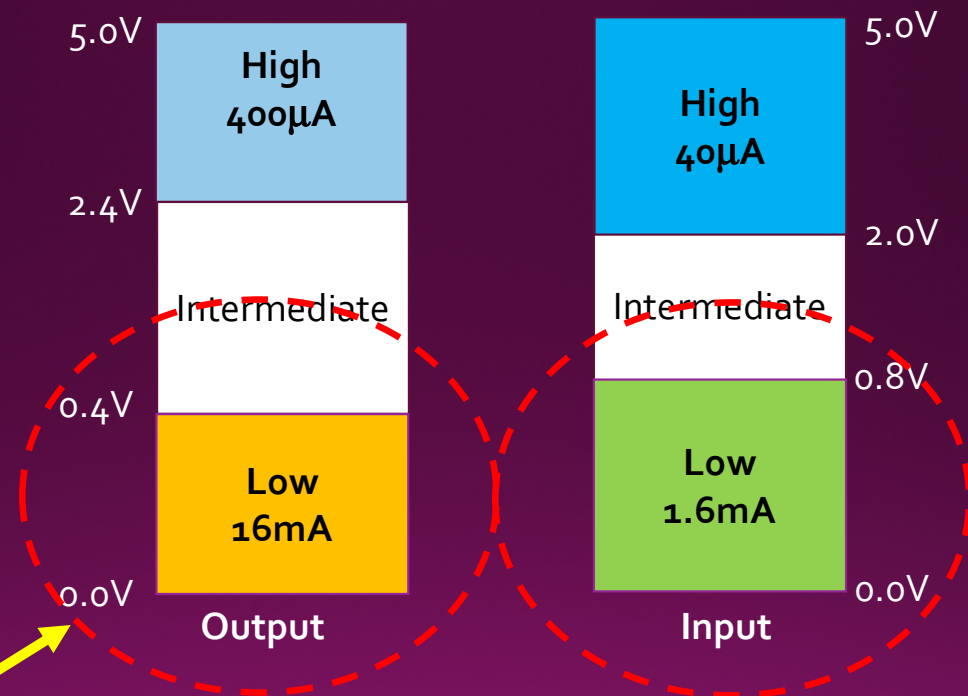


Standard TTL

Loading rules – Fan-Out



$$I_{IH(\text{max})} = 1.6\text{mA} \times 10 = 16\text{mA}$$



Standard TTL

Summarize

- Basics of digital Integrated circuits.
- Bipolar and MOS families.
- TTL – Transistor -Transistor Logic – Characteristics.
- TTL family.
- TTL Parameters.
- Working of TTL NOT/NAND gates.
- Floating inputs.
- Loading rules.

Thank You